YoungJo Choi (Jo Choi)

Digital Hardware Engineer

Brooklyn, NY | yc7330@nyu.edu | +1-312-918-9643

LinkedIn: linkedin.com/in/youngjo-choi-477787263 | GitHub: github.com/TechJoe96 | Portfolio: TechJoe96.github.io

EDUCATION

New York University Fall 2024–Present
- M.S., Computer Engineering (Expected graduation: May 2026)

Illinois Institute of Technology (Dual Degree) Fall 2022–Spring 2024
- B.S., Computer Engineering, Minor in Artificial Intelligence

Inha University (Dual Degree) Spring 2016–Spring 2024
- B.S., Electronic Engineering

SKILLS

- **HDL & Assembly:** SystemVerilog, Verilog, VHDL, RISC-V Assembly, ARM Assembly
- **Programming & Scripting:** Python, C/C++, MATLAB, Tcl
- **EDA & Design Tools:** Cadence Virtuoso/Genus/Innovus, Xilinx Vivado, ModelSim, OpenLane, KiCAD, AutoCAD
- **Design Methodologies:** RTL Design, UVM Verification, STA, FPGA Prototyping, ASIC Flow (Synthesis/P&R/Signoff)
- **Protocols & Interfaces:** DDR4 SDRAM, AXI4, Wishbone B4, SPI, I2C, UART
- **Other:** PyTorch, TensorFlow, Linux, Git, Docker

WORK EXPERIENCE

Data Scientist Intern at C&B Trading (Semiconductor Components), Seoul, Korea Jun 2024–Aug 2024
- Collaborated with the sales and operations teams to develop a time-series forecasting model in Python (pandas, scikit-learn) for semiconductor component demand, achieving 90% accuracy and replacing manual spreadsheet estimates
- Automated weekly data consolidation from order records, supplier price sheets, and shipping logs using Python (pandas), reducing analyst prep time by 60%
- Designed interactive PowerBI dashboards tracking supplier lead times, inventory turnover, and shipping delays across 3 regions, enabling management to renegotiate terms with 2 underperforming suppliers
- Analyzed pricing and margin data across 4 product categories, identifying optimal markup ranges that informed quarterly pricing strategy for cross-border transactions

RESEARCH EXPERIENCE & PROJECTS

DDR4-2400 SDRAM Memory Controller
- Designed a high-bandwidth memory controller compliant with the JEDEC DDR4-2400 standard, featuring bank-group-aware scheduling, 16 independent bank state machines, and automatic refresh with deferral to target 19.2 GB/s peak throughput
- Implemented the full memory initialization and data-transfer protocol, including power-on sequencing (MRS/ZQCL), read/write signal timing (DQS/ODT), and a CPU bus interface (AXI4) with burst support
- Validated design correctness using a behavioral memory model and automated checking framework (UVM scoreboard)—achieved 10/10 tests passed with zero data errors and zero timing violations

OpenPOWER SoC — Quantized Neural-Network Accelerator (QNNA)
- Designed a custom neural-network accelerator for on-chip ML inference, featuring an INT8 4×4 multiply-accumulate array with ReLU activation, integrated with the Microwatt OpenPOWER CPU via pipelined bus interface (Wishbone B4)—achieving 4× memory savings over FP32
- Developed RTL architecture with configurable matrix dimensions (M×N×K) and software-controlled computation via memory-mapped registers; wrote SystemVerilog testbench passing 6/6 SoC integration tests
- Taped out on SKY130 using the OpenLane automated ASIC flow (synthesis through signoff); generated final GDS-II on a 1×1 mm² die for the ChipFoundry Microwatt Design Challenge

Pipelined RISC-V CPU
- Designed a 5-stage pipelined CPU (fetch, decode, execute, memory, writeback) in SystemVerilog, implementing 40+ RISC-V integer instructions with hazard detection, data forwarding, and branch prediction
- Verified functional correctness using a self-checking testbench against a reference model, achieving ~1.2 cycles per instruction across arithmetic, branch, and memory-access benchmarks
- Synthesized for FPGA deployment at 100 MHz on Xilinx Artix-7 using Vivado; explored architectural trade-offs in branch prediction and cache hierarchy using the gem5 cycle-accurate simulator

Real-Time Parkinson's Tremor & Dyskinesia Detector
- Collaborated with 3 teammates to build a real-time wearable health monitor on STM32L475, detecting Parkinson's tremor and involuntary movement (dyskinesia) from motion sensor data using frequency analysis
- Implemented a digital signal-processing pipeline (DC removal, windowing, 256-point FFT) classifying tremor (3–5 Hz) and dyskinesia (5–7 Hz) at 52 Hz sampling rate with 95% detection accuracy
- Delivered working prototype with real-time visual feedback in 3-second analysis windows; presented results to faculty review panel as top-3 project in the course

ACTIVITIES & OTHERS

Teaching Assistant, New York University — ECE-UY 3114 Fundamentals of Electronics I Sep–Dec 2025

President & Founder, Korean Student Association at Illinois Institute of Technology Jun 2023–May 2024

Active Member, Triangle Fraternity (Armour Chapter) at Illinois Institute of Technology Sep 2022–May 2024

Republic Of Korea Military — Honorably discharged as Sergeant / Squad Leader (21 months) at 606 squad Aug 2017–Apr 2019